RTF: Real-Time Fingerprints Experimentation

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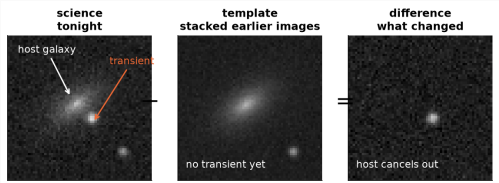
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The problem



- LSST will issue ∼10 million alerts every night.
- Each one: the light curve, a 63×63 pixel cutout triplet, and metadata.
- Brokers can only use a small subset, and the filtration leads to missed discoveries.

Inputs: the cutout triplet

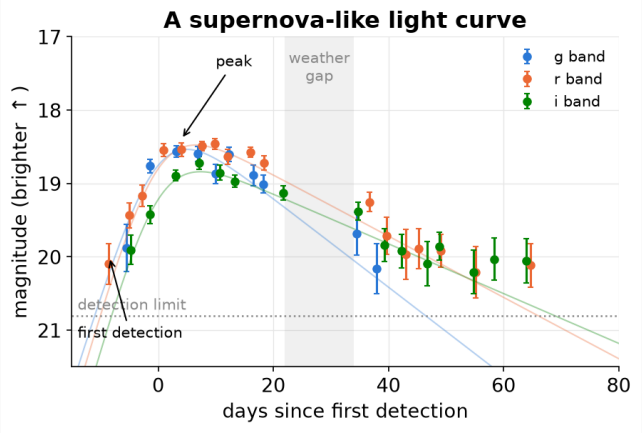


- **Science:** current observation, 63×63 pixels around the candidate. Captures the transient, host, neighbours, and the sky.
- **Template:** images of the sky before the transient.
- **Difference:** what changed between the template and science.

Why all three?

- The template is sharper than the science, so PSF scaling blurs the sharper template image before subtraction against the science.
- The template can give context about the host.
- Science + template can expose bad subtractions.

Inputs: the light curve



Brightness vs. time in three bands, sampled at uneven intervals when the survey records.

Datapoints:

time since first detection	$\log(1+\Delta t)$
gap since previous one	$\log(1+\delta t)$
flux, flux error	log
filter	g / r / i flag

Variable in length (depends on number of observations from the survey). **Switched off for this experiment.**

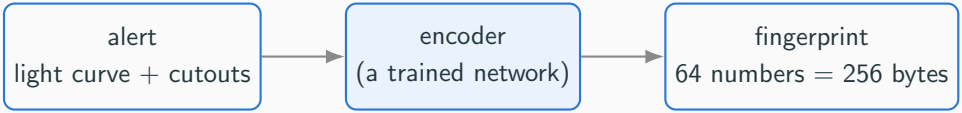
Inputs: the metadata

About 30 numbers the pipeline has already computed for every alert to give context. Examples:

- nearest catalogued source: distance, brightness, star or galaxy
- how point-like the detection is
- the survey's own real-vs-bogus score
- how many detections so far

Also switched off for this experiment.

The idea

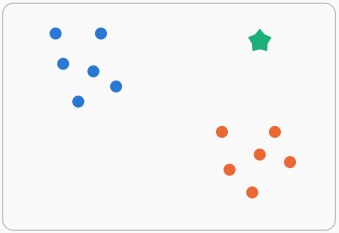


What we can do with fingerprints

- Supervised learning (classify).
- Unsupervised learning (cluster).

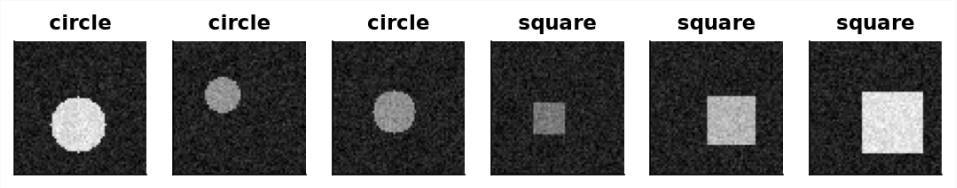
Solving for out-of-distribution





- The question we try to address: can an encoder trained on known classes also flag unknown ones?
- In other words: can we detect a new transient class without ever having seen one?

Simple experiment: circles vs. squares



- Light-curve input switched off: the fingerprint depends only on the cutout.
- Synthetic stamps: circles and squares of equal area, random size, position and brightness, plus noise.

Two ways to build the image encoder

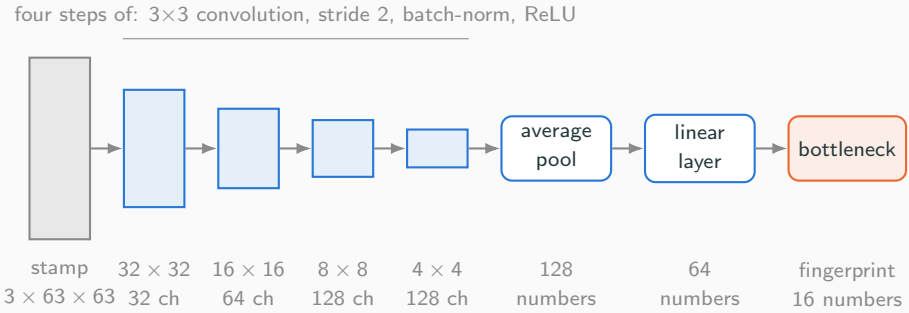
From scratch

- 0.25 million parameters
- starts knowing nothing
- learns only from the stamps

Zoobot (Walmsley et al.)

- 8.5 million parameters
- pre-trained on Galaxy Zoo
- frozen weights

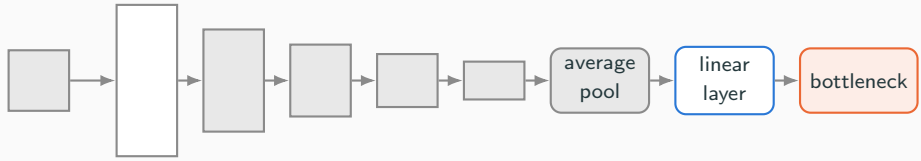
Encoder 1: the from-scratch CNN



- ∼0.25 million parameters, all trained on from random starting values using the stamps (blue = trained here).
- After pooling, the readout (linear + bottleneck) is identical for both encoders. Maps the pool of features to a 16-number fingerprint.

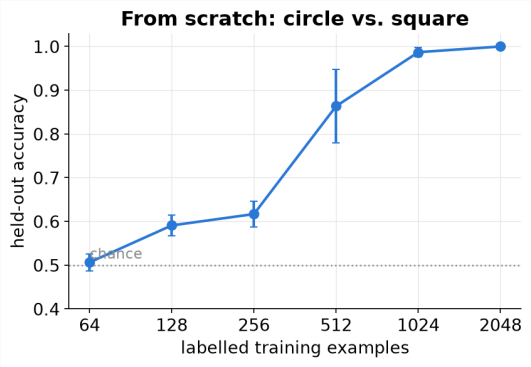
Encoder 2: Zoobot, frozen

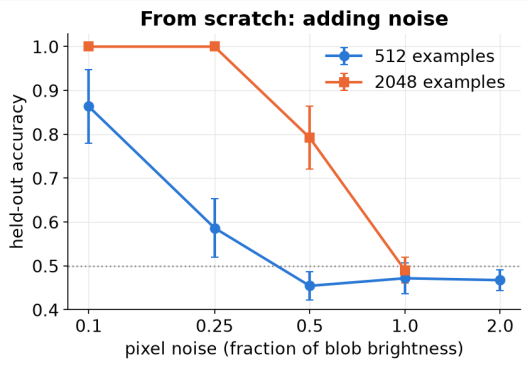
ConvNeXt-pico: four stages of 2, 2, 6, 2 blocks; Galaxy Zoo weights, **frozen**



- ∼8.5 million parameters, pre-trained to reproduce galaxy morphology on millions of galaxies. Grey = frozen, used exactly as published.
- The stamp is upsampled because the network expects 224×224 inputs.
- We train only the blue readout (∼35k parameters).

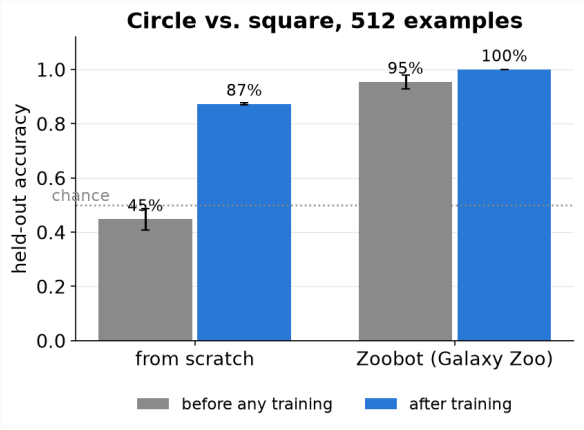
Result 1: from-scratch classification





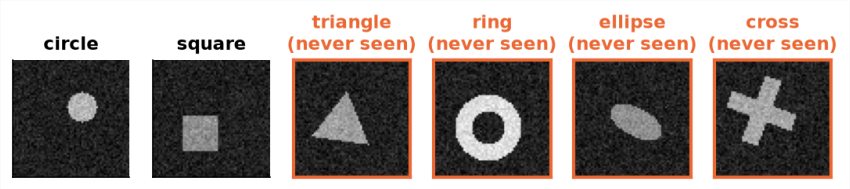
- 256 examples: 62 %. About 1000 examples: 99 %.
- Add modest noise and 512 examples is no longer enough, but 2048 fixes it.
- Takeaway: from-scratch compression works well, but requires lots of data (which may not exist for rare classes).

Result 2: Zoobot classification



- "Before" means a random readout after the frozen Zoobot encoder (zero training). "after" means the readout has been trained to classify circles vs. squares.
- Takeaway: Zoobot is already a capable encoder and requires far less data to accurately classify.

The discovery test: out-of-distribution shapes



- Train on circles and squares only. Then show four new shapes. We want to know:
  1. Are the fingerprint latents for the new shapes far from the existing circles and squares we trained on?
  2. If we group the fingerprint latents blindly, do the new shapes get their own groups?

Metrics used

New-shape detection

no labels used

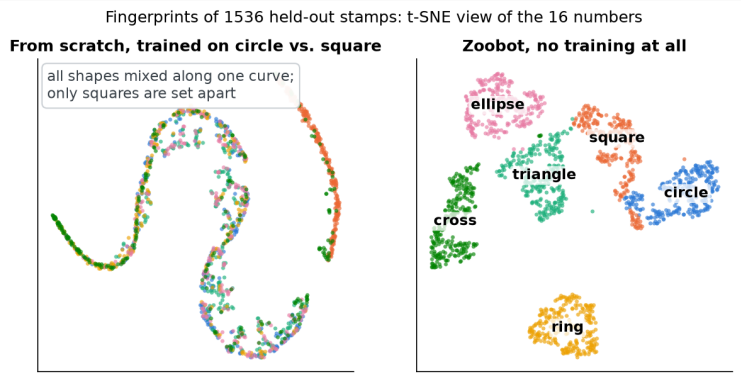
1. Encode the 2048 circle/square *training* stamps.
2. Encode the 1536 held-out stamps (256 per shape, six shapes).
3. For each held-out fingerprint, find its 5 nearest training fingerprints. The mean of those 5 distances is its novelty score.
4. We measure the chance that a random new-shape stamp outscores a random held-out circle or square (50 % useless, 100 % perfect).

Six-shape separability

labels used

1. Take the same 1536 held-out fingerprints, now with all six true labels.
2. Fit a 6-way logistic regression.
3. Accuracy = fraction of the 1536 predicted correctly. Chance $= 1/6 \approx 17\,\%$.
4. Determines if the class information is present in the fingerprint.

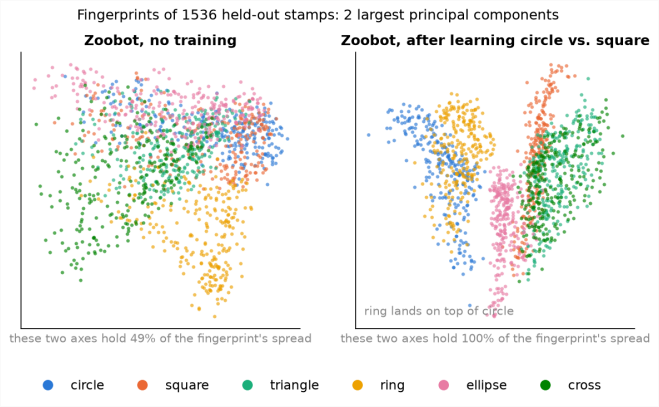
Result 3: from-scratch vs. Zoobot for new-shape detection



From scratch: new-shape detection 68 %, six-shape separability 43 %. Triangles, rings and ellipses are all grouped with circles. 1 effective dimension.

Zoobot, no training: detection 99.8 %, separability 99.5 %. New classes form their own islands. 5 effective dimensions.

Result 4: teaching Zoobot to classify collapses the fingerprint



- We fine-tune the Zoobot readout to classify circles vs. squares.
- Once the readout learns circle vs. square, the fingerprint collapses onto a line (1 effective dimension).
- Triangles, rings and ellipses are all projected onto the same line as circles.
- Six-shape separability: $99.5\,\% \rightarrow 85\,\%$.

Summary

Key takeaways

1. Use a Galaxy Zoo pre-trained encoder for cutouts. From-scratch isn't viable for rare class classification.
2. Pretraining substantially beats from-scratch for out-of-distribution.
3. A reconstruction-type objective is helpful during training to prevent collapse.

Limitations / Future directions

- The shapes in our experiment were easy, and real data will likely behave differently.
- We're still not sure if Galaxy Zoo specifically is producing the results, or if ordinary ImageNet-pretrained networks can achieve similar performance.
- Additional fine-tuning methods can be explored (e.g., unfreezing parts of the encoder).

Appendix

fingerprint built by	separability (with labels)	new-shape detection	blind grouping	dims used
untrained small network	0.23	0.55	0.00	3.3
small network, classify	0.43	0.68	0.28	1.0
small network, reconstruct	0.34	0.75	0.01	5.2
small network, both	0.44	0.68	0.23	2.7
Zoobot, no training	**0.995**	**0.998**	0.54	4.8
Zoobot, classify	0.85	0.96	0.54	1.0
Zoobot, both	0.97	0.97	**0.62**	1.4

2048 training stamps, 256 held-out stamps per shape.